Mohamed Amine Bouzahar

Senior Full-Stack & AI Engineer · 12 years of experience

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PROFILE

I build production backends and full-stack platforms in Python, TypeScript and Java, and I run them on Kubernetes. Since 2024 I have led applied-AI delivery: RAG pipelines, MCP servers and agent tooling that moved internal assistants from prototypes to tools teams use every day.

Delivered for: European Space Agency · Hermès · Foncia · SNCF Connect · BPCE · Vodafone · Eutelsat · Telecentro · Treemetrics

EXPERIENCE

Senior Full-Stack & AI Engineer — Infogene (Emeria / Foncia)

05/2026 – Present

- Rebuilt legacy services as RabbitMQ-driven microservices with Temporal orchestrating long-running flows, so product domains release independently.
- Set up the MCP servers and agent rules the team uses daily, giving every pull request an automated first review.
- Built a native macOS control centre and a headless JSON CLI over the same guarded engine, so engineers and AI agents drive the dev environment through identical operations.
- Audited the team's AI tooling spend and halved the cost of long investigations by changing how sessions are run.

`TypeScript · NestJS · React · MongoDB · RabbitMQ · Temporal · GraphQL · GitLab CI · Datadog`

Lead Full-Stack, AI & DevOps Engineer — Valeuriad

09/2024 – 05/2026

- Led the Python and AI workstream across the internal platforms and built the company's first RAG pipeline, so the assistant answers from the company's own documents.
- Shipped the FastAPI services behind the AI assistant and took it from prototype to company-wide use.
- Moved internal projects to Kubernetes on GCP with GitHub Actions and Azure DevOps pipelines and SonarQube quality gates.
- Implemented SAML 2.0 and Google SSO, and mentored the team through knowledge-transfer sessions.

`Python · FastAPI · SQLAlchemy · Pytest · Java · Quarkus · RAG · MCP · Angular · Kubernetes · GCP · Azure DevOps`

Full-Stack Engineer, Python / React — SNCF Connect (consultant via Valeuriad)

01/2025 – 08/2025

- Delivered Python backends and AWS Lambda microservices for internal operations tooling.
- Built the React interfaces those teams work in daily and added the RobotFramework suite that removed the manual regression pass.

`Python · React · Material UI · Vite · AWS Lambda · PostgreSQL · DynamoDB`

Senior Full-Stack Engineer & DevOps — Capgemini (Hermès)

04/2022 – 09/2024

- Built the AWS Lambda and SNS/SQS/SES pipelines that process notification and order flows asynchronously.
- Moved deployments to ArgoCD GitOps and built the Grafana dashboards that caught incidents before customers did.
- Connected applications to Okta SSO and ran the technical interviews that decided who joined the team.

`AWS Lambda · SNS / SQS / SES · ArgoCD · Grafana · Okta SAML · Symfony · Akeneo PIM`

Software Engineer — Wiztivi

02/2019 – 04/2022

- Delivered set-top-box interfaces shipped to Vodafone, Eutelsat and Telecentro subscribers, tuned for constrained hardware.
- Wrote the Python tooling that automated set-top-box detection and debugging for the whole team.

`TypeScript · JavaScript · Python · Grafana`

Full-Stack & IoT Engineer — Tisalabs

07/2017 – 12/2018

- Built the Flask REST APIs behind a monitoring application for the European Space Agency and Treemetrics.
- Trained TensorFlow models and built a vector-embedding search on Elasticsearch, years before it came off the shelf.
- Designed MQTT sensor ingestion and a RabbitMQ messaging prototype.

`Python · Flask · TensorFlow · Elasticsearch · RabbitMQ · MQTT · Docker`

Web Developer & Designer — Freelance (Morocco)

12/2014 – 06/2017

- Ran end-to-end web projects for educational associations and local businesses: requirements, database design, data migration, deployment and client training.

`SQL Server · MySQL · PHP`

SKILLS

Backend	Python, FastAPI, SQLAlchemy, Pydantic, Pytest, Node.js, NestJS, Java (Spring Boot, Quarkus)
AI / LLM	RAG pipelines, vector search, MCP servers, LangGraph multi-agent workflows, hosted and self-hosted models
Frontend	TypeScript, React, Angular, Material UI, Vite, Nx monorepos
Messaging	RabbitMQ, Temporal, AWS SQS / SNS, MQTT, event-driven architectures
Data	PostgreSQL, SQL Server, MySQL, MongoDB, DynamoDB, ETL
Security	SAML 2.0, Okta, PingID, OAuth 2.0 / OIDC, JWT, RBAC
DevOps	Kubernetes, Helm, ArgoCD, Docker, Terraform, GitHub Actions, GitLab CI, Azure DevOps, AWS, GCP
Practices	Hexagonal architecture, domain-driven design, tests first, code review, mentoring

OPEN SOURCE & PERSONAL PROJECTS

- **mdns-listener-advanced** — Zero-config mDNS / Zeroconf discovery and publishing of .local hostnames, now on its fourth major version.
- **ngx-toast · react-toast** — One notification library shipped as two framework-native builds: zoneless-ready for Angular, hook-based for React.
- **advanced-color-utils** — Colour manipulation and accessibility toolkit.
- **Prayer times platform** — FastAPI and PostgreSQL backend with solar-position astronomy implemented from scratch, a React client generated from its OpenAPI schema, shipped as a Helm chart on my own Kubernetes cluster.
- **iOS & watchOS app** — Offline Qibla compass, prayer times and Quran reader: iPhone app, widgets and an Apple Watch companion, in Swift 6 and SwiftUI.
- **canvas-floor-maps** — Canvas-based floor plan rendering library.

EDUCATION

- **Master's degree in Computer Engineering** — EHEIO, Oujda (2012 – 2017)
- **Bachelor in Mathematics & Computer Science** — Université Jean Monnet, Saint-Étienne (2008 – 2012)

LANGUAGES

French (Native) · English (C2) · Arabic (Native) · Tamazight (Native) · German (A1) · Dutch (A0 · learning)